

The HippoCamera Knowledge Hub: A digital educational program for older adults to promote healthy aging

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- HippoCamera (HC) is a mobile application designed to improve memory via cue creation.¹
- Many older adult users of HC were curious about the science behind the app. To address this, we developed an educational program in the app to equip older adults with the knowledge necessary to stave off cognitive decline.^{2,3}
- Such a tool offers a tangible way which older adults can use to strengthen their cognition.

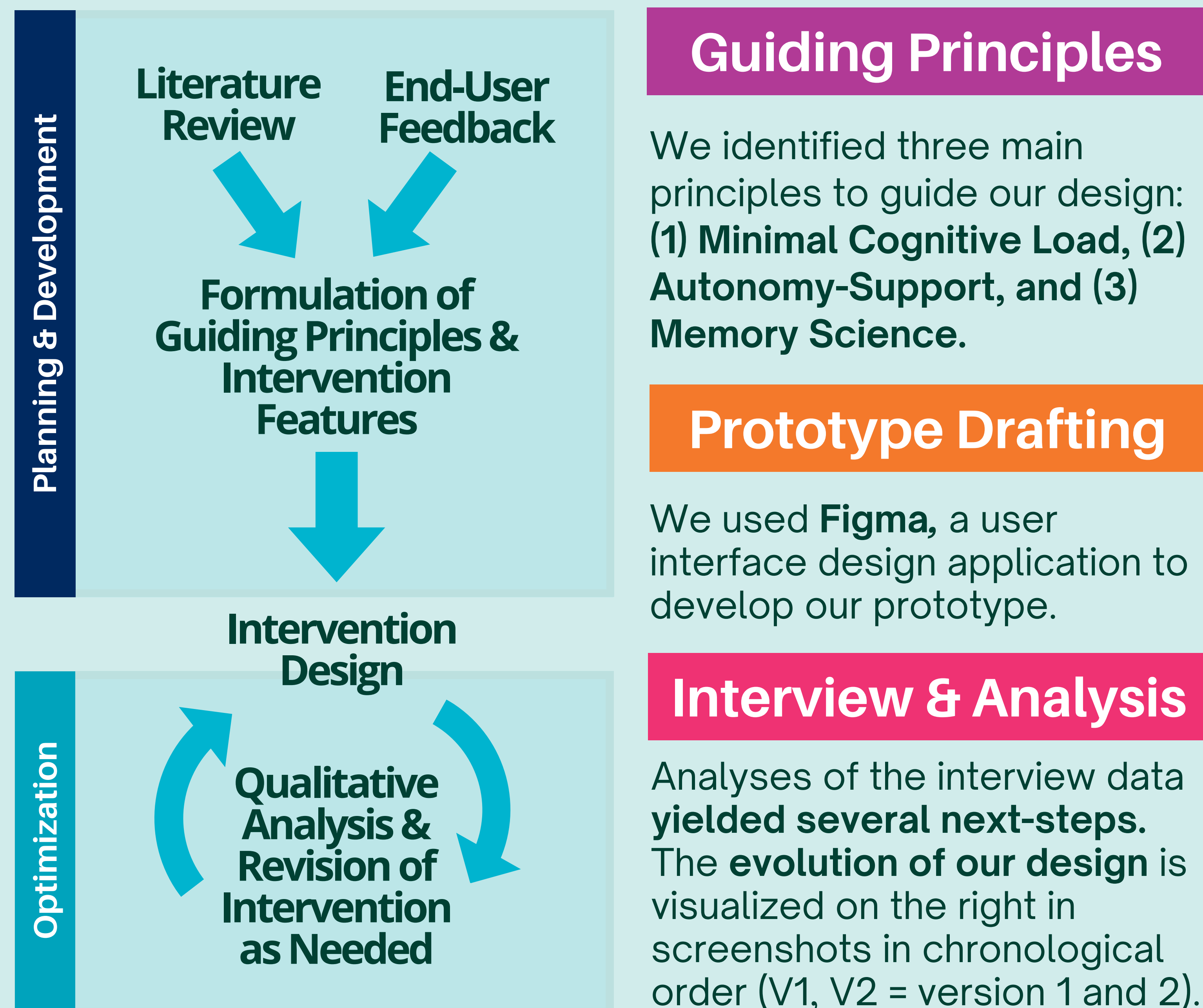
How do we build an educational program for older adults that leverages the memory and implementation sciences?

We interviewed **12 older adults** ($M_{age} = 75.75$; $SD = 4.14$; 1M/10F; 11 v1/1 v2) from the Greater Toronto Area as they used our prototype. We assessed usability via think-aloud and semi-structured interviews, the **net promoter score (NPS, out of 10)**, and the **modified mHealth App Usability Questionnaire (mMAUQ, out of 7)**.⁶

NPS: $M = 6.50$ ($SD = 1.98$) | **mMAUQ:** $M = 5.01$ ($SD = 1.28$)

The Person-Based Approach^{3,4}

The person-based approach (PBA) is a framework that integrates the perspectives of end users throughout the development process, while also drawing from existing evidence and theory.^{3,4}

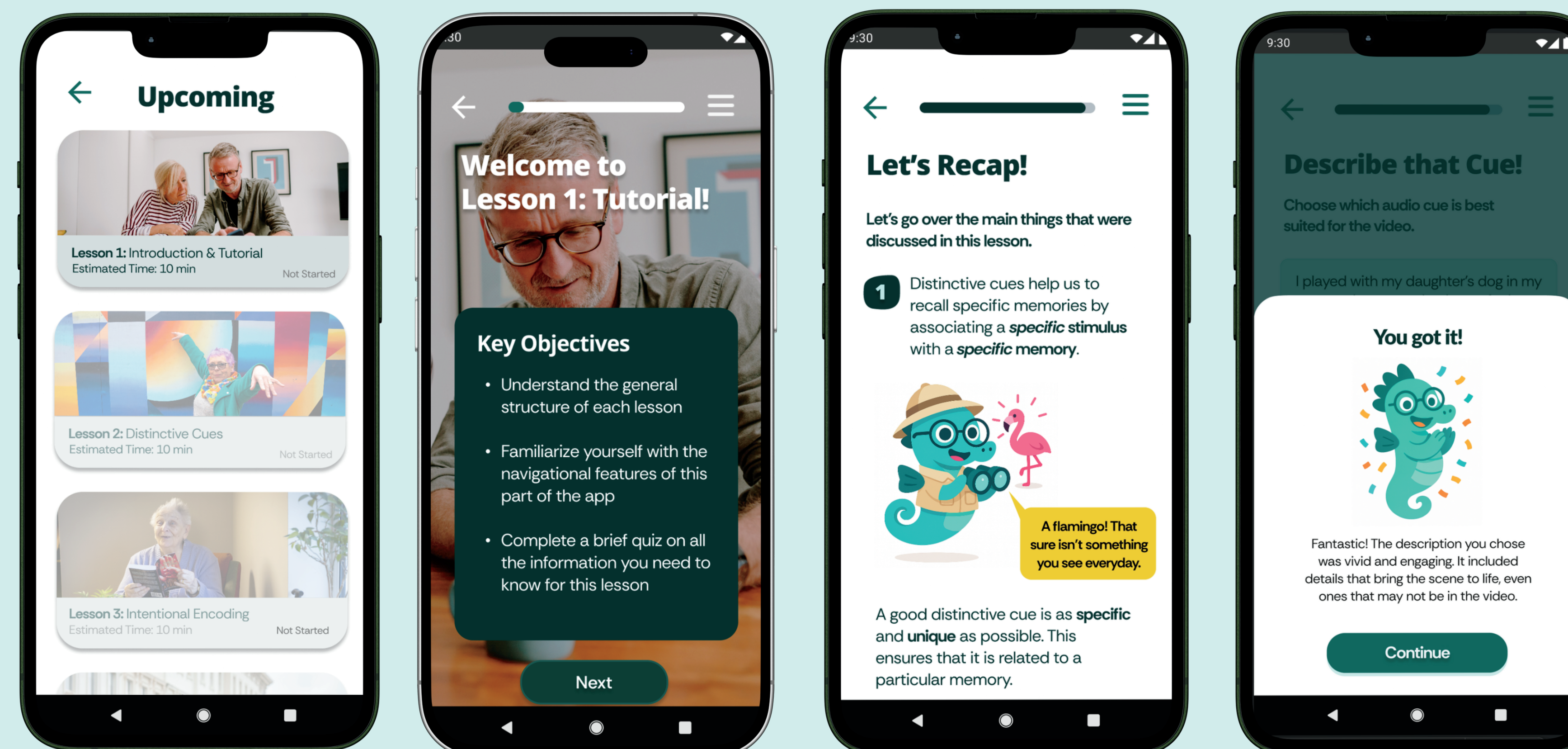


TDF⁷, BCT⁸, COM-B⁹

Future Directions

- Use behavioural change techniques to analyze qualitative data.⁸
- Continue to **explore and apply feedback on future iterations** of the current prototype.

Prototype V1



Adding Topics

"....[there should be] some psychoeducation on healthy aging...and reassuring people that...memory loss is normal." (P10)

Tutorial Removal

"This is sort of redundant. 'What to expect from the memory strategies module?' Well, I'm here to improve my memory." (P04)

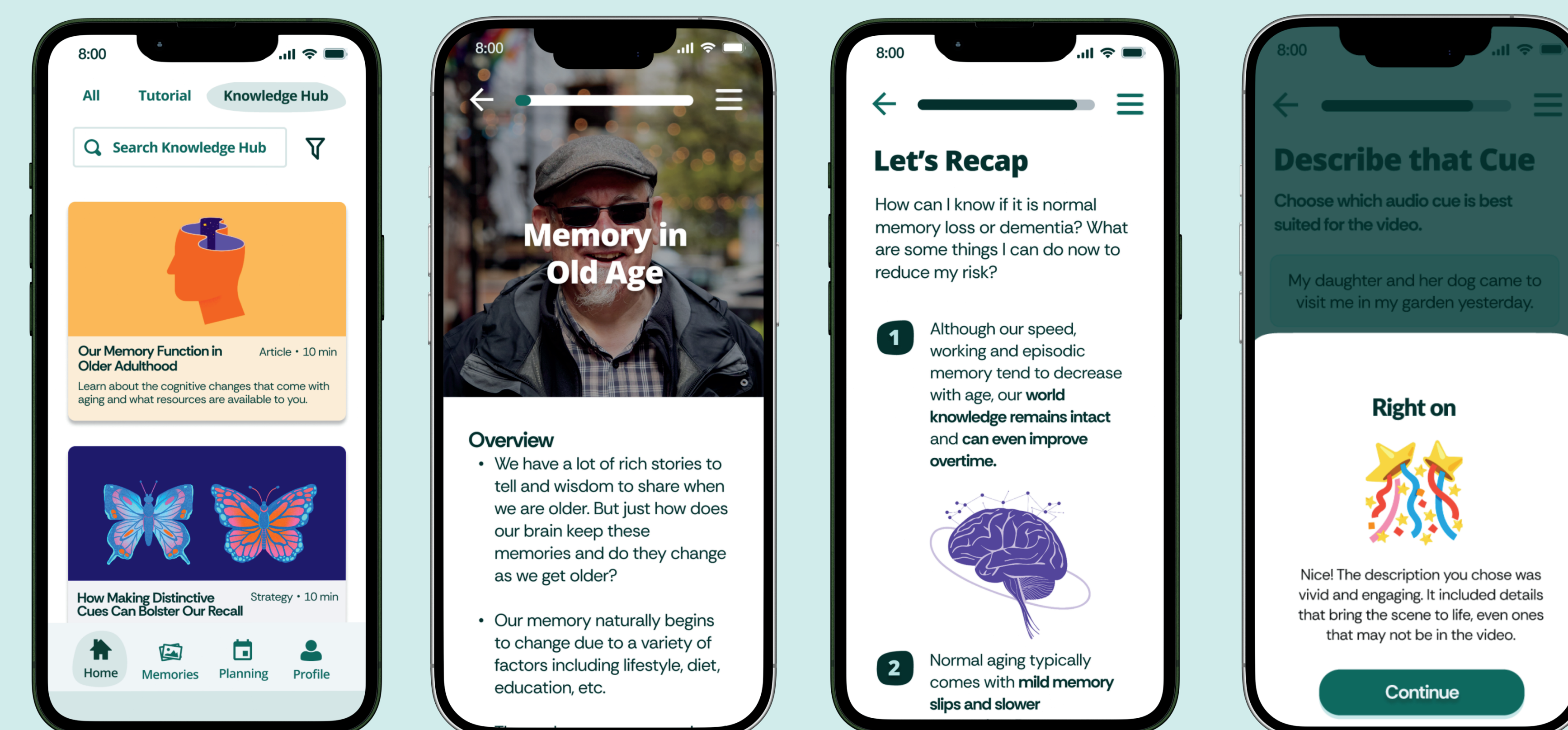
Mascot Removal

"It's a...completely different vocabulary than this cartoon character...You're going from a childish thing to this very high tech thing." (P02)

Language Change

"...use of exclamations is a little bit overdone. 'That's it for our first lesson! *exclamation mark*'... really treating seniors [like] they're children." (P11)

Prototype V2



Summary of Design Changes

- Introduction of more relevant examples
- Less prescriptive language
- Integration of topics into a "Knowledge Hub"
- "Blog/Article" format versus *lesson* format
- Non-cartoonish graphics

Version 2 Data

So far, the second version of our prototype with these changes was **received very positively**. Our participant indicated **several strengths** with the design including: relevance of examples, ease-of-use, and intriguing graphics.

"...I mean, I'm already thinking just trying to use what I've learned here as soon as I leave this building." (P12)

"...I felt engaged and helpful, useful, and also made me feel like I am normal..." (P12)

References: ¹Martin et al. (2022). *PNAS*, 119(51), e2214285119. ²Yardley et al. (2015). *J Med Internet Res*, 17(1), e4055. ³Yardley et al. (2015). *Pilot Feasibility Stud*, 1(1), 1-7. ⁴Reeve et al. (2021). *Educ Psychol*, 56(1), 54-77. ⁵Zhou et al. (2019). *JMIR Mhealth Uhealth*, 7(4), e11500. ⁶Braun, V., & Clarke, V. (2006). *Qual Res Psychol*, 3(2), 77-101. ⁷Michie et al. (2013). *Ann Behav*, 46(1), 81-95. ⁸Atkins et al. (2017). *Implement Sci*, 12(1), 1-18. ⁹McDonagh et al. (2015). *Implement Sci*, 13, 130.